

Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Different Techniques To Calculate Needs For Your Project



Learning Outcome:

Implement and document the software tests, applying trial and error, based on the designed plan and procedures established by the company.

Applies the different types of software testing to the IT solution.

Ask your partner:

- What are the difficulties that people may face when calculating time needed to finish software testing?
- Do you think it is easy to meet the deadlines of a project? Why?

Can Do Statements:

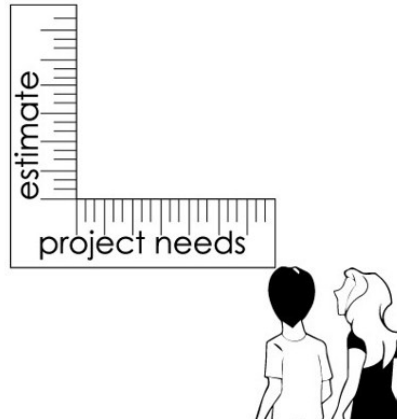
- I can explain what the work breakdown (WBS) structure estimating technique is.
- I can describe how the three point estimation works.
- I can talk about the functional point method to estimate.





Let's Explore!

Kiara and Thomas are overviewing some topics related to software testing estimation.



Instructions: Read the text below.

- Kiara:** Hi, Thomas! How are you doing with this estimation techniques topic?
- Thomas:** I'm doing well, but if you want, we could practice together a little bit. I need to explain the different estimation methods which people use to plan a project. I'm afraid of failing the test.
- Kiara:** Not likely to happen, but let's practice anyways. Tell me about the Work Breakdown Structure.
- Thomas:** In the WBS, the project is divided into detailed steps to pinpoint mistakes more easily. Since they share the responsibility of handing in assignments on time, this structure allows the group work members to foster commitment toward the assigned tasks.
- Kiara:** Perfect! Now, tell me about the three-point estimation method that project leaders use.
- Thomas:** Well, when the project is scaffolded, three estimations are made. First, an optimistic one; where everything works perfectly and nothing bad ever happens. Secondly, the pessimistic one, where everything goes wrong. And third, the realistic one, that offers the most possible scenario for the project weighing the pessimistic and optimistic scenarios.
- Kiara:** Not bad! Ok, so what's the functional point method about?
- Thomas:** That one really drives me crazy! Well, the project is divided into functional points. Then, each part is assigned a weighting, which is later multiplied by category, and then the total is multiplied by the rule of 2 person-hour, which gives the final estimation for the project.
- Kiara:** That is a tricky one, but you got it right. Remember that you also need to take into account the size of the project and each of its parts. Statistical data is also a very important tool that helps you optimize resources and create a wider or tighter schedule.
- Thomas:** Thanks Kiara, this review will come in handy during my test!

**Let's Practice!****Pessimism, Optimism and Reality**

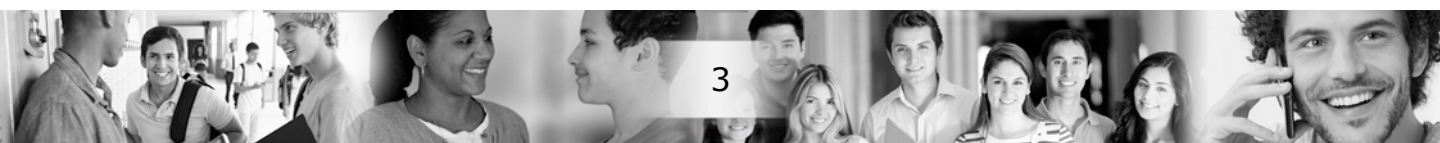
Instructions: Organize the mystery sentence. Rewrite the sentence on the line provided.

1. made. / the / when / is / scaffolded, / Well, / three / are / project / estimations
2. is / estimation / an / optimistic / is / where / There / perfectly / done. / everything
3. executed / Anything / in / an / estimation. / is / optimistic
4. are / pessimistic / done / wrongly / a / Things / estimation. / in
5. and / scenarios / realistic / Pessimistic / are / weighed / optimistic / in / a / estimation.

**Let's Practice!****The Project's Division**

Instructions: Select the correct answer.

1. The functional point method is
 - a. the one that divides the project into different points.
 - b. the one that merges the project into different points.
 - c. the one that divides the project into the same points.
 - d. the one that unites the project into different points.
2. In this functional method, each point is assigned a weighting which is later
 - a. multiplied by category.
 - b. divided into the same categories.
 - c. summarized in each category.
 - d. multiplied by the same categories.
3. The total is multiplied by the rule of _____ which gives the final estimation for the project.
 - a. 5 person-hour
 - b. 2 person-hour
 - c. 3 person-hour
 - d. 4 person-hour
4. Remember that
 - a. you don't need to take into account the size.
 - b. you don't need to take into account the size of the project.
 - c. you need to take into account the size of the project.
 - d. you do need to take into account the size of the categories.
5. Statistical data is
 - a. a very important tool that helps you optimize resources.
 - b. a not so important tool that helps you optimize resources.
 - c. a very important tool that helps you optimize categories.
 - d. a very important tool that helps you reduce resources.





Extension Activity

Let's practice some techniques!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Design a brochure where you describe the different estimating techniques.
- In the space you provide for each technique specify their characteristics.
- Use relative clauses and simple present passive.
- Share the information with your class.

Self-Assessment

Yes

No

- I design an element to describe the different estimating techniques
- I know the characteristics of each estimating techniques.
- I use relative clauses and simple present passive correctly
- I talk about estimating techniques.

☐ ☐
☐ ☐
☐ ☐
☐ ☐


Target Words

- **pinpoint** (verb): to show where something is.
e.g. In the WBS, the project is divided into detailed steps to pinpoint mistakes more easily.
- **foster** (verb): to promote or impulse.
e.g. Since they share the responsibility of handing in assignments on time, this structure allows the group work members to foster commitment toward the assigned tasks.
- **hand in** (verb): to give or submit a task.
e.g. Since they share the responsibility of handing in assignments on time, this structure allows the group work members to foster commitment toward the assigned tasks.
- **scaffolded** (adjective): to be divided into small steps.
e.g. Well, when the project is scaffolded, three estimations are made.
- **weighting** (noun): a value in a scale given to something.
e.g. Then, each one is assigned a weighting, which is later multiplied by category, and then the total is multiplied by the rule of 2 person-hour, which gives the final estimation for the project.
- **come in handy** (expression): something considered to be useful.
e.g. Thanks Kiara, this review will come in handy during my test!



Learning Points

Subject and object relative clauses

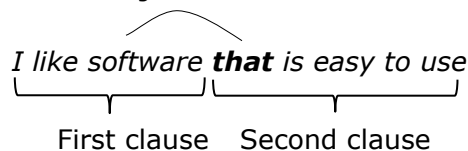
Subject and object relative clauses help you shorten the sentences and avoid redundancy.

Relative clauses are used to link two sentences that share information about the same subject or object. They help to avoid repetition of pronouns and make sentences better structured.

The **defining relative clauses** present relevant information to understand the whole sentence and **non-defining relative clauses** present extra information that is not relevant.

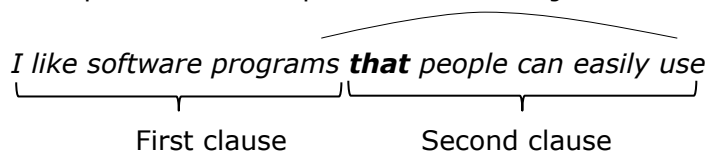
The relative pronouns we use to link sentences are: **"who" (people)**, **"which" (objects)** and **"that" (people and objects)**.

Relative pronouns can define the subject or object of the clause. Let's check some examples of relative pronouns as **subjects** of relative clauses.



In this example, the relative pronoun is working as the subject of the next clause. We can replace **"that"** with **"software"** and in that way we know it is the subject of the second clause.

Now let's check an example of a relative pronoun as the **object** of the clause:



In this example, the relative pronoun is linking two sentences but it works as the object because the effect of the verb **"use"** relies on **"software programs"**. In this case, **"software programs"** is not performing the action of the second clause. Notice that in this case, there is a clear subject in the second clause.

Examples

- *I need to explain the different estimation methods which people use to plan a project.*
- *Statistical data is also a very important tool that helps you optimize resources and create a wider or tighter schedule.*





Learning Points

Estimating techniques

There are different **estimation techniques** that allow people to calculate the human, physical, and monetary resources needed for testing software.

The first estimation technique is known as WBS or **Work Breakdown System**, where people divide the project in smaller steps, making it easier to spot mistakes and correcting them. Each member of the team is in charge of a piece of the project, once it is revised, it is handed in and shared with the other members of the team so a better feedback is provided.

The second estimation method is called the **Three-point Estimation**, which is done after the work breakdown process. Once the project has been divided, team members make three estimations. The first estimation is **optimistic**, since they visualize the best possible scenario where all conditions are favorable and beneficial. The second estimation is **pessimistic**, where all problems and issues pop out and completing the task seems to be farther every time. The last one is in between, the **realistic** estimation, where a more real scenario takes place, some issues may appear but nothing that hinders the process significantly.

The last estimation method is the **functional point**. This one requires a little bit more of detail. The first step is to divide the project into functional points (referring to what the end-user will see on the screen once it's finished). Then, every functional point is assigned a weighting (value) that ranges from 1 being simple, 3 being medium, to 5 being difficult. Next, the weighting is multiplied by the number of functional points in the category. This gives a total which is later multiplied by the norm of **2 person-hours** and that final result gives the estimation of the entire project.

Those three estimation methods are the most common and offer a "reliable" estimation. Of course, any estimation will demand tons of practice because measurement is often done subjectively.